

XRPL Validator — Daily Engineering Report • 2026-08-10

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 6 commits (xrpl-validator 6)

Engineering changes

- **fix(xrpl-ledger): reprice bridged leg A after the mid-leg rounds up** (07f2e32 , xrpl-validator) 9bb5ceb rounded the bridged XRP mid-leg UP to whole drops and left `gets_in` priced from the FRACTIONAL value it had before. So leg A took a whole drop and paid for a fraction less of one, every time the rounding did anything.
- **fix(xrpl-ledger): a payment pays the INPUT issuer's TransferRate** (c99ebb2 , xrpl-validator) `apply_path_payment` charged the OUTPUT issuer's rate (`want_rate`) and let the input through free. A payment sets `ownerPaysTransferFee_ = false`, so `trIn = transferRate(book_.in.account)`: the SENDER parts with `amount × rate` and the counterparty receives `amount` (BookStep.cpp:737-739). `apply_iou_direct` has always done this for the same-currency case — the crossing path never did.
- **fix(xrpl-ledger): match rippled's AMM slice sizing on direct offer crossing** (b2c47da , xrpl-validator) Two coupled rules, both read off libxrpl's own trace of #106143011 4607B92B against the RLUSD/XRP pool and confirmed in rippled's source.
- **fix(xrpl-node): choose the hydration RPC per LEDGER, not per sweep script** (25c9e31 , xrpl-validator) The gate corpora AGE OUT of our LAN validator's rolling window (~290k ledgers on 2026-08-10, 105919185-106209950), so a fixture that hydrated locally last month must come from s2 today. A server pinned per script cannot express that, and both failure directions were live: * `batch2sweep.sh pins --rpc <lan>` while 7 of its 26 ledgers had rolled out of the window (105909285 105909624 105910649 105911834 105912291 ...
- **fix(xrpl-ledger): mint real LP tokens for AMMCreate and single-asset deposits** (85bb7df , xrpl-validator) Both mint sites were hardcoded 1e7 placeholders, documented as "the magnitude is oracle-corrected downstream". They were 23 of the 83 value divergences at gate 46 and by far the largest in the backlog — relative errors up to 3e5, where the whole remaining OfferCreate tail sits below 2e-4.
- **fix(xrpl-ledger): quantize a deposit's LP mint to the pool balance's ulp** (acf49ab , xrpl-validator) SETTLES THE MANTISSA-SCALE QUESTION, which 85bb7df left open as the suspected cause of the single-asset deposit precision tails. It is NOT the cause.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **MPC Ceremony Attestation on SignerListSet** (custody) Add SFields sfMPCCeremonyHash (Hash256) and sfMPCQuorum (t-of-n) plus a tfMPCAttested flag to SignerListSet; when set, the transactor requires the has
- **Vault NAV Marking for Tokenized Funds** (rwa) Add sfNAVPerShare (Number) and sfNAVManager (AccountID) SFields to the SingleAssetVault ledger object, and a new VaultMarkNAV transaction, signable on
- **Iceberg Offers with TickSize-Quantized Clips** (dex) Add a tfIceberg flag to OfferCreate plus an sfDisplayQuantity SField (the visible clip, required to be a multiple of the trading pair's TickSize preci

Appendix — commits

Time	Repo	Commit	Subject
01:37	xrpl-validator	07f2e32	fix(xrpl-ledger): reprice bridged leg A after the mid-leg rounds up
10:25	xrpl-validator	c99ebb2	fix(xrpl-ledger): a payment pays the INPUT issuer's TransferRate
13:03	xrpl-validator	b2c47da	fix(xrpl-ledger): match rippled's AMM slice sizing on direct offer crossing
20:36	xrpl-validator	25c9e31	fix(xrpl-node): choose the hydration RPC per LEDGER, not per sweep script
22:24	xrpl-validator	85bb7df	fix(xrpl-ledger): mint real LP tokens for AMMCreate and single-asset deposits
23:23	xrpl-validator	acf49ab	fix(xrpl-ledger): quantize a deposit's LP mint to the pool balance's ulp